Exact Obstructions, Exact Structure, and Certified Bounds for the Three-Dimensional Ising Model

Exact 3D Ising Research Program

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Abstract

The three-dimensional nearest-neighbour Ising model is not solved, and nothing below claims to solve it. This paper consolidates what an exact, machine-audited research program has actually established: a family of independent no-go theorems that close named routes to solvability, exact characteristic-zero structure theory for the Lie algebras generated by layer transfer terms, and an exactly certified computational base of series coefficients and rigorous critical bounds.

The obstructions fall into five logically independent classes. *Local-algebraic*: a closed-form Dolan–Grady defect that vanishes exactly when every layer vertex has degree 0 or 2, and a tridiagonal (Askey–Wilson) no-go for representative three-dimensional layers. *Representation-theoretic*: exact characteristic-zero decompositions of the two-generator layer algebras at 2×3, 2×4 and 3×3, which combined with the Date–Roan classification of Onsager-algebra quotients show that these algebras are not Onsager quotients in any disguise; and an exact classification of the local-term algebras of open $2 \times L$ ladders, $\mathfrak{so}_8^2$, $\mathfrak{sp}_{32}^2$, $\mathfrak{so}_{128}^2$, $\mathfrak{sp}_{512}^2$, now proved for *every* $L \geq 2$, which yields an unconditional isotropic all-L free-fermion no-go with exponentially growing mode floors. *Spectral and basis-independent*: exact certificates that layer transfer spectra are not Gaussian subset-product multisets, culminating in the open 2×6 layer, where a two-prime pair-product computation gives $\deg \gcd \leq 295414 < 295415$ and hence at least one distinct pair product above the twelve-mode Gaussian ceiling; plus an all-size anisotropic theorem with a constant-fraction excess. *Conserved-quantity*: the only finite-support operator on $\mathbb{Z}^3$ commuting with the mixed generator is a scalar, and the Pauli-string centralizer of the layer generating set is $\{I, \prod_i X_i\}$. *Tensor and planarity*: the identical-factor tetrahedron equation has only the zero auxiliary matrix away from $q \in \{0, \pm 1\}$, the positive-real scalar Kac–Ward cone is empty, and 31 of 32 rational Kac–Ward branches are empty over $\mathbb{Q}$.

The computational base is exact: the simple-cubic reduced free energy is computed to v^{28} at high temperature and x^{56} at low temperature with per-box Chinese-remainder uniqueness certificates, and the critical coupling is certified to lie in [0.2122159753270231627267517174278577806020, 0.2527310098586630
Every statement carries an explicit status tag, every universal claim is separated from the finite computation that motivated it, and the withdrawn claims are listed as prominently as the proved ones.

Contents

1 Introduction

1.1 What is claimed and what is not

The two-dimensional Ising model is solvable because several distinct structures coincide: the transfer generators satisfy the Dolan–Grady relations and generate the Onsager algebra [11, 9, 10], the local terms map to Majorana bilinears [28], the model is planar so the Kac–Ward determinant applies [19], and the spectrum is a free-fermion subset-product multiset [24, 20]. In three dimensions each of these structures fails, but "fails" is a mathematical assertion that must be proved for a precisely stated mechanism, not asserted from the absence of a known solution. Computational hardness results [18] say that no efficiently computable partition function exists for general three-dimensional instances; they do not by themselves exclude a closed form for the translation-invariant simple-cubic model.

This paper states and proves the failures, one mechanism at a time, with exact arithmetic and with scope sentences that are meant to be adversarial against their own conclusions. What is *not* claimed anywhere: a thermodynamic-limit free energy, magnetisation, correlation function, spectrum, exact critical coupling, or exact critical exponent. No result below excludes an exact solution based on a Gaussian restriction of a larger system, non-Pauli or quasilocal conserved quantities, matrix-valued or dynamical tensor ansatzes, nonlocal transformations, or a mechanism not yet formulated.

1.2 Status tags

Every mathematical statement in the underlying repository, and every statement here, carries one of:

- **[THEOREM]** — proved in the stated scope, which is always finite, algebraic, or explicitly universal with a proved induction.
- **[LEMMA]** — proved auxiliary statement.
- **[COMPUTATION]** — an exact finite calculation with a stored, independently replayable artifact. A finite computation is never promoted to an all-sizes theorem without a proved induction.
- **[EXTERNAL]** — established elsewhere and cited, used as input or as a post-computation witness, never as a fitting target.
- **[UNRESOLVED]** — honestly open, including cases where a resource wall was measured.

A measured timeout is never a theorem. The benchmark value $K_c = 0.221654626$ [14] is used for comparison only; it is never used to fit, select, tune, or validate a derivation.

1.3 Principal results

Table 1 lists the load-bearing results with their scopes.

2 Setup and conventions

2.1 Model and transfer representation

The model is the ferromagnetic nearest-neighbour Ising model on the simple cubic lattice, $H = -J\sum_{\langle xy\rangle}\sigma_x\sigma_y$ with $\sigma_x \in \{\pm 1\}$, $K = \beta J$, $v = \tanh K$, $x = e^{-2K}$ and $u = x^2$. The reduced free energy is normalised as in the repository specification: at high temperature $\phi(v) = \log 2 + \sum_n f_n v^n$, and at low temperature $\phi(K) = 3K + \sum_{n\geq 1} c_n x^n$.

Transfer is taken along one axis, so that each layer is a two-dimensional graph $\Gamma = (V, E)$ and the layer-to-layer operator is, up to a positive scalar,

$$V_\Gamma = e^{K^* A} e^{KB}, \qquad A = \sum_{v\in V} X_v, \qquad B = \sum_{(ij)\in E} Z_i Z_j, \tag{1}$$

result	content	scope
Dolan–Grady defect (§3.1)	closed-form residual $R(\Gamma)$; vanishes iff every degree is 0 or 2	every finite simple graph
Tridiagonal no-go (§3.2)	no Askey–Wilson relation; one-line certificate	eight named layers, affine-invariant
Claw obstruction (§3.3)	term-wise Majorana bilinearity forces $\Delta(\Gamma) \leq 2$	every finite graph
Local commutant (§3.4)	only scalars commute with $a\sum X + b\sum ZZ$	all sizes, finite support, $ab \neq 0$
Pauli centralizer (§3.4)	centralizer $= \{I, \prod X_i\}$	connected graphs, Pauli group
Onsager-quotient no-go (§4)	exact char-0 types at 2×3, 2×4, 3×3; not Onsager quotients	three named layers
$2 \times L$ classification (§4.2)	$\mathfrak{so}_8^2, \mathfrak{sp}_{32}^2, \mathfrak{so}_{128}^2, \mathfrak{sp}_{512}^2, \ldots$; equality proved	every $L \geq 2$
Spectral no-gos (§5)	non-Gaussian spectra at 2×3 to 2×6, 3×3	named couplings, then coupling-generic
Theorem F (§5.5)	all-size excess $48\cdot 3^{n-4}$ above the Gaussian ceiling	anisotropic, off a hypersurface
Tetrahedron no-go (§6.1)	only $R=0$ for $q \notin \{0, \pm 1\}$	identical factors, 8×8 auxiliary
Kac–Ward (§6.2)	positive-real cone empty; 31/32 branches empty over $\mathbb{Q}$	scalar translation-invariant family
Exact series (§7)	v^{28} high-temperature, x^{56} low-temperature	exact rational, CRT-certified
Certified interval (§7.3)	$K_c \in [0.21221597\ldots, 0.25273100\ldots]$	rigorous, not sharp

Table 1: Load-bearing results and their exact scopes.

with X, Y, Z the Pauli matrices on $(\mathbb{C}^2)^{\otimes|V|}$ and K^* the dual coupling defined by $e^{-2K^*} = \tanh K$ [28]. Writing $t = \tanh(K^*/2)$ and $q = (1+t^2)/(2t) = e^{2K}$ makes V_Γ an exact rational matrix at rational t, which is what allows every spectral statement below to be made with no floating point.

The single structural feature that distinguishes three dimensions is elementary and decisive: a one-dimensional layer graph has $\Delta(\Gamma) \leq 2$, whereas a genuine cubic layer contains a vertex of degree at least three. Several theorems below are exactly the statement that some solvability mechanism is equivalent to $\Delta(\Gamma) \leq 2$.

2.2 Two different algebras

Two Lie algebras appear throughout and must not be confused.

Definition 2.1. The *two-generator algebra* is $\langle A, B\rangle_{\text{Lie}}$, generated by the two sums in (1). The *local-term algebra* is $\mathfrak{g}_\Gamma = \langle X_v\ (v \in V),\ Z_iZ_j\ ((ij) \in E)\rangle_{\text{Lie}}$, generated by the individual terms.

For the open 2×3 layer the two-generator algebra has dimension 263 and the local-term algebra has dimension 1056; at 2×4 the dimensions are 2952 and 16256. Statements about one are not statements about the other, and modular witnesses computed for one have been mis-attributed to the other in earlier drafts of this program. Both algebras are relevant, for different mechanisms: the two-generator algebra is the object of the Onsager/Dolan–Grady route, while the local-term algebra is the object of term-wise free-fermion routes [6, 12, 5].

2.3 Exact-arithmetic protocol

Three conventions carry the weight of every numerical claim below.

Two primes, one direction. Ranks and greatest-common-divisor degrees computed modulo a prime bound the characteristic-zero quantity in exactly one direction: a modular rank is a lower bound for the rational rank, and a modular gcd degree of a monic-normalised pair is an upper bound for the characteristic-zero gcd degree. Agreement at two primes is a cross-check, never a proof of rational equality. Substituting a bound into the wrong side is the single error class that has cost this program a theorem, and one withdrawn result (§8) is exactly that mistake.

Monic specialization.

Lemma 2.2 (Monic-in-parameter upgrade)**.** *Let $F(z;\tau) \in \mathbb{Z}[\tau][z]$ be monic in z, and let $G = \gcd(F, \partial_z F)$ computed over the fraction field. Then for every specialization $\tau \mapsto \tau_0$, $\deg\gcd(F(\cdot;\tau_0), \partial_z F(\cdot;\tau_0)) \geq \deg G$. Consequently one evaluated point bounds the generic value, and the resultant $\rho = \mathrm{Res}_z(F/G, \partial_z F/G)$ confines the exceptional parameters to the finite zero set of a single nonzero polynomial.*

This lemma is what converts finite point certificates into coupling-generic theorems, and it is applied only in the direction it permits.

Verification independence. Every deliverable consists of a producer script, a standalone verifier that re-derives the decisive quantity from raw inputs without importing the producer's conclusions, an artifact with content digests, and a tagged note. Cross-artifact references digest the semantic `data` section with sorted keys rather than whole-file bytes, because timestamp refreshes otherwise break dependent tests. The full acceptance gate is 123 standalone verifiers, all passing.

3 Local-algebraic obstructions

3.1 The Dolan–Grady defect

The Dolan–Grady relations

$$[A,[A,[A,B]]] = 16[A,B], \qquad [B,[B,[B,A]]] = 16[B,A] \tag{2}$$

generate the Onsager algebra and are the algebraic core of the two-dimensional solution [11, 9].

Theorem 3.1 (Dolan–Grady defect; [THEOREM])**.** *For every finite simple graph Γ, with $d(v) = \deg(v)$ and neighbourhood $N(v)$, the first relation in (2) holds identically, and the second has the exact residual*

$$R(\Gamma) = [B,[B,[B,A]]] - 16[B,A] = 24i \sum_{v\in V} Y_v\Big((d(v)-2)\sum_{u\sim v} Z_u + 2\sum_{\{u,u',u''\}\subseteq N(v)} Z_u Z_{u'} Z_{u''}\Big). \tag{3}$$

Hence $R(\Gamma) = 0$ if and only if $d(v) \in \{0,2\}$ for every v; the quartic part of $R(\Gamma)$ vanishes if and only if $\Delta(\Gamma) \leq 2$; and the number of four-body Pauli terms is exactly $\sum_v \binom{d(v)}{3}$, the number of induced claws in the frustration graph.

Proof sketch. Expand the nested commutators term by term in the Pauli basis. Only edges incident to a selected vertex v contribute, because $[Z_i Z_j, X_v] = 0$ unless $v \in \{i,j\}$, and every surviving term carries the factor Y_v at that single vertex. Distinct vertices therefore cannot cancel one another, and the coefficient bookkeeping at a fixed v gives the degree-linear term $(d(v)-2)\sum_{u\sim v} Z_u$ and the neighbour-triple term. Both coefficients are integers, so $R(\Gamma)=0$ forces every local coefficient to vanish, which for $d(v)\geq 3$ is impossible because the triple sum is nonempty and its Pauli strings are linearly independent of the linear ones. □

Two points deserve emphasis. First, the isolated-vertex case matters: the frequently stated equivalence "Dolan–Grady holds iff Γ is 2-regular" is false, since $d(v)=0$ also annihilates the defect; a triangle plus an isolated vertex is the decisive counterexample. Second, the quartic operator $Y_v Z_{u_1} Z_{u_2} Z_{u_3}$ produced by a degree-three vertex is the same operator that obstructs the tridiagonal family in §3.2, which is why those two no-gos, though logically independent, always fail together.

Corollary 3.2 ([COMPUTATION])**.** *The identity (3) was verified term by term, with exact integer coefficients, on sixteen graph families including paths, cycles, stars, complete graphs, trees, square grids, a torus layer, and a cubic graph.*

Simultaneous deformation was also tested and closed: within the invariant supports F_1–F_4, no simultaneous nonlinear deformation of both generators restores (2), and the finite F_3 deformation that cancels the second relation on the 3×3 torus provably fails the first. The two varieties are simultaneously empty for the minimal orbit basis, with Gröbner basis $\{p^2, pq, q^2, pr, qr, r^2\}$ and all saturations trivial.

3.2 Tridiagonal and Askey–Wilson relations

The tridiagonal condition, the structure behind the Onsager algebra and the Askey–Wilson relations, is tested through

$$T_0 = \beta T_1 + \gamma^* T_2 + \rho^* T_3, \qquad (T_0, T_1, T_2, T_3) = ([B, B^2A + AB^2], [B, BAB], [B, BA + AB], [B, A]),$$

which is linear in the three unknown parameters at fixed A, B.

Theorem 3.3 (Representative tridiagonal no-go; [THEOREM])**.** *For the boundaryless torus layers 3×3, 3×4, 4×4, 3×5, 4×5, 5×5, and for the open 2×3 and 3×3 layers, the second tridiagonal relation has no solution over $\mathbb{Q}$, hence none over $\mathbb{C}$. The conclusion is invariant under every affine redefinition $A' = aA + s$, $B' = bB + t$ with $ab \neq 0$.*

For the 4×4 torus a single Pauli coefficient is a complete certificate:

$$W = Z_0 Y_5 Z_6 Z_{10}, \qquad [T_0]_W = -48, \qquad [T_1]_W = [T_2]_W = [T_3]_W = 0.$$

No universal one-row witness exists; some geometries require the stored four-row certificate, and claiming otherwise was an early over-generalisation in this program.

3.3 Term-wise Majorana bilinearity and the claw

The frustration graph of the standard generating set is the subdivision $S(\Gamma)$: each X_i anticommutes exactly with the bond terms incident to i, while the X terms commute among themselves and the bond terms commute among themselves.

Theorem 3.4 (Claw obstruction; [THEOREM])**.** *If a set of Hermitian Pauli terms is mapped injectively, after a unitary change of basis, to Majorana bilinears $i\gamma_p\gamma_q$, then its frustration graph contains no induced $K_{1,3}$. Hence the standard layer generating set admits a term-wise Majorana-bilinear representation only if $\Delta(\Gamma)\leq 2$. Every square-grid layer with a vertex of degree three or four fails.*

Proof. Two bilinears anticommute exactly when their two-element index sets meet in one element. Suppose a bilinear b_0 has three pairwise commuting neighbours b_1, b_2, b_3 in the frustration graph. Each b_k shares exactly one index with b_0, and b_0 has only two indices, so two of the b_k share the same index with b_0; but then those two share an index with each other, so they anticommute, contradicting that they commute. Hence no induced claw. □

This is the graph-invariant criterion of the free-fermion-solvability literature [6, 12, 5], specialised to the Ising layer; note that free-fermion solvability can survive in "disguised" form for other generating sets [13], which is exactly why the spectral tests of §5 are needed.

A quantitative companion covers enlarged Gaussian realizations.

Proposition 3.5 (Mode-count bound; [THEOREM]). *If the generated real Lie algebra $\mathfrak{g}$ embeds in the span of bilinears of m fermionic modes, i.e. $2m$ Majorana operators, then*

$$\dim \mathfrak{g} \leq \dim \mathfrak{so}(2m) = m(2m-1), \qquad \textit{so} \qquad m \geq \frac{1+\sqrt{1+8\dim\mathfrak{g}}}{4}.$$

On the physical Hilbert space of a layer with n sites one has $m = n$, so any exact lower bound on $\dim \mathfrak{g}$ exceeding $n(2n-1)$ excludes a same-space Gaussian realization outright. This is a dimension obstruction; it does not exclude an embedding into a strictly larger space.

3.4 Conserved quantities

Theorem 3.6 (Pauli centralizer; [THEOREM]). *Let Γ be connected and $G = \{X_i : i \in V\} \cup \{Z_i Z_j : (ij) \in E\}$. The Pauli strings commuting with every element of G are exactly $\{I,\ \prod_{i\in V} X_i\}$, and the same holds for the summed pair $\{A, B\}$ within the Pauli group.*

Proof. Encode a string as $(a \mid b) \in \mathbb{F}_2^{2n}$. Commuting with every X_i forces $b = 0$; commuting with every bond forces $a_i = a_j$ along each edge. Connectedness leaves $a = 0$ or $a = (1, \ldots, 1)$. □

This closes the Kitaev-style family of same-space Pauli fluxes, and only that family: non-Pauli symmetries, enlarged-space constructions, and ordinary lattice translation sectors are outside its scope, and translation on a triangle is an explicit counterexample to the stronger statement that was once claimed here.

The genuinely all-size statement is about finite-support operators on the infinite lattice.

Theorem 3.7 (Local commutant; [THEOREM]). *Fix a, b with $ab \neq 0$ and let $H_g = a\sum_x X_x + b\sum_{\langle xy\rangle} Z_x Z_y$ on $\mathbb{Z}^3$. In the finite-commutator-derivation sense, the only compact finite-support operator commuting with H_g is a scalar. The simultaneous A/B centralizer is a corollary.*

Proof sketch. Take a support site v with maximal first coordinate and its private outside neighbour $w = v + e_1$. Matching the Z_w slot of the commutator forces $b\,[Q, Z_v] = 0$-type structure; matching the Y_v slot then forces $2ia\,Q_Z = 0$; support induction removes the site and repeats. Both generator coefficients are used, which is why $a = 0$ or $b = 0$ is excluded. □

Explicitly outside scope: extensive translation sums of local densities, quasilocal or infinite-support charges, and the finite-torus operator $\prod_x X_x$. The extensive class was treated separately: for translation-covariant $Q = \sum_x \tau_x(q)$ with finite density q, the derivation vanishes iff the local commutator is a lattice divergence, so the nontrivial quotient dimension is $\operatorname{rank} S_R - \operatorname{rank} M_R - 1$. The one-dimensional chain gives exactly $2R$ for $R = 0, \ldots, 9$, with the oriented energy current already at $R = 1$, while $\mathbb{Z}^2$ at $R = 1, 2$ and $\mathbb{Z}^3$ at $R = 1$ give exactly 1: only the Hamiltonian density survives. In the $\mathbb{Z}^3$ radius-two box the support-size class $s \leq 5$ is decided exactly — 21,121,156 columns, rank $S = 14{,}757{,}412$ and rank $M = 14{,}757{,}410$ at both primes 2147483647 and 2147483629, meeting the analytic upper bound rank $S-2$, so the quotient is exactly 1. The general absence of local conserved quantities in $d \geq 2$ for this model class is external [7]; the finite-shape reconstruction here is an independent exact computation, not a rival proof.

4 Exact characteristic-zero structure of layer algebras

4.1 The two-generator algebras and the Onsager-quotient no-go

Failure of Dolan–Grady for the displayed generators does not preclude the generated algebra from being an Onsager-algebra quotient in some disguise. That loophole is closed by computing the algebra exactly and comparing with the classification of Onsager quotients [8, 29].

Theorem 4.1 (Exact char-0 structures; [THEOREM]). *Over $\mathbb{Q}$, for the two-generator algebras of the open layers,*

$$\begin{aligned}
2\times 3:&\quad \dim = 263 = 1 + \left[C_7 + C_3 + C_3 + A_8 + A_5\right], \quad \textit{Killing rank } 262,\\
2\times 4:&\quad \dim = 2952 = \mathbb{Q}z + \mathfrak{so}(F_{000}) + \mathfrak{so}(F_{010}) + \mathfrak{so}(F_{100}) + \mathfrak{sl}_{24} + \mathfrak{sl}_{28} + 2\,\mathfrak{sl}_4,\\
&\quad \textit{absolute type } D_{21} + 2B_{13} + A_{23} + A_{27} + 2A_3, \quad \textit{radical} = \textit{centre},\\
3\times 3:&\quad \dim = 8034 = \mathbb{Q}z + \mathfrak{sl}_{33} + \mathfrak{sl}_{48} + \mathfrak{sl}_{59} + \mathfrak{sl}_3 + \mathfrak{sl}_{16} + \mathfrak{sl}_{30}, \quad \textit{type } A_{32} + A_{47} + A_{58} + A_2 + A_{15} + A_{29}.
\end{aligned}$$

At 2×3 the C_7-versus-B_7 ambiguity is resolved by an exact invariant alternating form of determinant 16384. Each decomposition was obtained by two independent exact-$\mathbb{Q}$ closures, and a clean-room verifier that imports nothing from the producer re-derives the controlling minors.

Corollary 4.2 (Not an Onsager quotient; [THEOREM] + [EXTERNAL]). *The Levi subalgebra of any finite-dimensional quotient of the Onsager algebra is a sum of copies of $\mathfrak{sl}_2$ [8]. None of the three algebras above has that Levi type. Hence the 2×3, 2×4 and 3×3 layer algebras are not Onsager-algebra quotients in any disguise: no generating pair of them satisfies Dolan–Grady.*

The corollary is stated for the three named layers, because it rests on exact computations for those layers. For the open 4×4 layer only $\dim_{\mathbb{Q}} \mathfrak{g} \geq 1794$ is proved, by 1794 stored literal Pauli words; its exact dimension, type, centre, radical and Killing rank are open, blocked by a 268591168-dimensional container.

4.2 The $2\times L$ alternation family

The local-term algebras of open ladders form the most structured family found so far. Let $n = 2L$, let $J = (1,\dots,1) \in \mathbb{F}_2^n$, let d be one checkerboard class of the ladder, and set

$$S_L = \{(x \mid z) \;:\; |z| \text{ even},\; z \neq J,\; (J-z)\cdot x = 1 + z\cdot d\}, \tag{4}$$

together with the full fibre $\{(x \mid J) : x \in \mathbb{F}_2^n\}$ when L is odd.

Lemma 4.3 (Uniform forward closure; [LEMMA]). *For every $L \geq 2$ the set S_L is closed under every nonzero adjoint action of an X_i or a ladder bond Z_uZ_v. Hence $\mathfrak{g}_{2\times L} \subseteq \mathrm{span}_{\mathbb{Q}}(S_L)$.*

Proof sketch. Three cases. An X_i action preserves the affine fibre defining (4). A bond action preserves the law because every ladder bond crosses the checkerboard class. The exceptional case $z + z_g = J$ either has vanishing commutator, when L is even, or lands in the full $z = J$ fibre, when L is odd. $\square$

Theorem 4.4 (All-L containment; [THEOREM])**.** *For every $L \geq 2$, with $n = 2L$, and with respect to the signed checkerboard form Ω_L on the two $\prod_i X_i$ parity halves,*

$$\mathfrak{g}_{2\times L} \subseteq \begin{cases} \mathfrak{so}_{2^{n-1}}(\mathbb{Q}) \oplus \mathfrak{so}_{2^{n-1}}(\mathbb{Q}), & L \text{ even}, \\ \mathfrak{sp}_{2^{n-1}}(\mathbb{Q}) \oplus \mathfrak{sp}_{2^{n-1}}(\mathbb{Q}), & L \text{ odd}, \end{cases} \qquad \dim \mathfrak{g}_{2\times L} \leq 2^{n-1}(2^{n-1} - (-1)^L).$$

The alternation is forced by the identity $\Omega_L^T = (-1)^L \Omega_L$, and membership in the form is the same parity equation as the set law, $|x| \equiv 1 + z \cdot x + z \cdot d \pmod 2$.

Theorem 4.5 (All-L classification; [THEOREM])**.** *Equality holds in Theorem 4.4 for* every *$L \geq 2$. Writing $m = 2^{n-1}$ with $n = 2L$,*

$$\mathfrak{g}_{2\times L}(\mathbb{Q}) = \begin{cases} \mathfrak{so}_m(\mathbb{Q}) \oplus \mathfrak{so}_m(\mathbb{Q}), & L \text{ even}, \\ \mathfrak{sp}_m(\mathbb{Q}) \oplus \mathfrak{sp}_m(\mathbb{Q}), & L \text{ odd}, \end{cases} \qquad \dim_{\mathbb{Q}} \mathfrak{g}_{2\times L} = 2^{n-1}(2^{n-1} - (-1)^L).$$

Proof sketch. The reverse inclusion is proved by an explicit production system on Pauli labels. Encode a monomial as $(a \mid b) \in \mathbb{F}_2^{2n}$ with symplectic pairing $\langle (a \mid b), (c \mid e) \rangle = a \cdot e + b \cdot c$; the generators are $(e_v \mid 0)$ for sites and $(0 \mid e_u + e_v)$ for bonds, and the single production rule is that p, q reached with $\langle p, q \rangle = 1$ implies $p + q$ reached, which is exactly the statement that a nonvanishing Pauli bracket lies in the algebra. The induction carries an explicit invariant $I(L)$ and uses two independent moves: a length-two route across one square, which yields all four pure-X three-corner labels, and a square-triple second move. A transport lemma shows a complete pair fibre moves along a two-token edge with no rank loss, the bond filter contributing $D_{\text{source}} \cap \ker(h)$ while the site term at the newly added endpoint supplies the missing direction; a leaf induction establishes token-graph connectivity including the boundary columns $c = 1$ and $c = L$; complete pair fibres then generate every odd pure-X label and every higher even-z fibre. This is an all-size argument, not an extrapolation. □

The exact types and dimensions at small L are

L	n	$\dim \mathfrak{g}_{2\times L}$	exact type over $\mathbb{Q}$
2	4	56	$\mathfrak{so}_8 \oplus \mathfrak{so}_8$
3	6	1056	$\mathfrak{sp}_{32} \oplus \mathfrak{sp}_{32}$
4	8	16256	$\mathfrak{so}_{128} \oplus \mathfrak{so}_{128}$
5	10	262656	$\mathfrak{sp}_{512} \oplus \mathfrak{sp}_{512}$
6	12	4192256	$\mathfrak{so}_{2048} \oplus \mathfrak{so}_{2048}$

Each algebra is perfect and splits into two ideals selected by the $\prod_i X_i$ parity halves; at $L = 3$ the diagonal Killing signature is $(544, 512)$ with 528-dimensional ideals, and at $L = 4$ the signature is $(8192, 8064)$. Independent confirmation: exact integer breadth-first closure enumerates 262656 labels at $L = 5$ and 4192256 at $L = 6$ with zero set-law violations, the latter in 18.6 process seconds at 458 MB peak resident size, and the production/rank construction covers all 8191 ordinary even-z support rows at $L = 7$, giving 67117056. The $L = 7$ figure is *not* a full enumeration and is labelled as such in the artifact.

Remark 4.6 (Structural cause of the alternation)**.** The $\mathfrak{so}/\mathfrak{sp}$ alternation is not an accident of the checkerboard bookkeeping. A second, independent route makes it classical. Order the n sites along the perimeter Hamiltonian path and apply Jordan–Wigner, introducing $2n$ Majoranas. Every site term and every path bond becomes a Majorana bilinear, and consecutive bilinears generate all of $\mathfrak{so}(2n)$; each remaining rung becomes a single Clifford monomial of grade $2(q-p)$, which bipartiteness forces to be $\equiv 2 \pmod 4$. Since a grade-k monomial reverses with sign $(-1)^{k(k-1)/2}$, which is -1 exactly for $k \equiv 2 \pmod 4$, the span of that grade class is the (-1)-eigenspace of Clifford reversal, i.e. the algebra preserving the spinor bilinear form on each half-spin space — symmetric when $n \equiv 0 \pmod 4$ and alternating when $n \equiv 2 \pmod 4$. With $n = 2L$ that is exactly L even versus L odd, and the dimension count agrees identically:

$$\sum_{\substack{0\le k\le 2n \\ k\equiv 2\ (4)}} \binom{2n}{k} = 2^{n-1}(2^{n-1} - (-1)^L), \qquad n = 2L.$$

Theorem 4.5 combined with Proposition 3.5 gives an unconditional all-L obstruction, with no genericity and at the physical isotropic point.

Corollary 4.7 (All-L quadratic no-go; [THEOREM]). *For every $L \ge 2$ the $2L$ site terms and $3L-2$ bond terms of the open $2 \times L$ layer cannot all be Majorana-quadratic, in any basis of the physical 2^n-dimensional space: already $\dim \mathfrak{g}_{2\times L} \ge n(2n-1)+1 > \dim \mathfrak{so}(2n)$. Quantitatively, any exact quadratic realization on an invariant code needs $m \ge \lceil (1+\sqrt{1+8D_L})/4 \rceil$ modes, and for $L \ge 3$ $m > 2^{2L}/\sqrt{5(4L+1)}$, so the mode count grows exponentially in L; the exact conservative floors at $L = 2, \dots, 8$ are $6, 18, 65, 216, 991, 3168, 15354$.*

The $L = 2$ case is worth isolating, because the claw obstruction of Theorem 3.4 does not apply there: the 2×2 ladder is a 4-cycle with maximum degree two, yet $\dim \mathfrak{g}_{2\times 2} = 56 > 28 = \dim \mathfrak{so}(8)$, forcing $m \ge 6$. Two scope limits are explicit: the enlarged-space statement assumes exact quadratic intertwining on an invariant code, so a non-invariant compression is not covered, and this is a statement about the algebra generated by the local terms, not about the spectrum of a single transfer matrix. The corresponding two-generator bounds are separate: $\dim\langle A_L, B_L\rangle \ge K_L - W_L$ with $K_L = (2^{2L-1} + 3 \cdot 2^L)/4$ and $W_L = 0, 2, 10, 66, 364, 1822$ for $L = 3, \dots, 8$, giving 6562 against the ceiling $2L(4L-1) = 496$ at $L = 8$, and no all-L law for W_L is yet proved.

4.3 A general trichotomy: paths, cycles, and branching

The Clifford-grade mechanism behind Theorem 4.5 is not specific to ladders, and pushing it to general graphs produces the sharpest structural statement in this paper. It also produces an exact refutation of the naive generalisation, which is recorded first because it is what fixes the hypotheses.

Let Γ be a finite simple bipartite graph on n vertices possessing a Hamiltonian path, let $N = 2n$, and let $\mathfrak{g}_\Gamma$ be the local-term algebra acting on the full 2^n-dimensional space. Number the path positions $0, \dots, n-1$ and Jordan–Wigner in that order. A chord joining positions $p < q$ becomes a single Clifford monomial of grade $2(q-p)$, and bipartiteness forces $q - p$ odd.

Lemma 4.8 (Branching equivalence; [LEMMA]). *For a graph carrying a Hamiltonian path the following are equivalent: some chord is not the pair of path endpoints; some chord has grade $k \le N-4$; and $\Delta(\Gamma) \ge 3$.*

Proof. Every interior path vertex already has path degree two, so a chord other than the endpoint pair meets an interior vertex and raises its degree to at least three. Conversely if $\Delta \le 2$ no chord meets an interior vertex, so Γ is the path or the path plus its endpoint edge, i.e. a cycle. The endpoint edge has path distance $n-1$, grade $N-2$; every other chord has distance at most $n-2$ and grade at most $N-4$. □

Theorem 4.9 (Exact trichotomy; [THEOREM])**.** *With* Γ*,* n*,* N *as above, exactly one of the following holds.*

branch	*generated Clifford grades*	$\dim_{\mathbb{Q}} \mathfrak{g}_\Gamma$
Γ *is the path*	$\{2\}$	$\binom{N}{2} = n(2n-1)$
Γ *is an even cycle*	$\{2,\, N-2\}$	$2\binom{N}{2} = 2n(2n-1)$
$\Delta(\Gamma) \geq 3$	*every* $k \equiv 2 \pmod 4,\ k \neq N$	$\begin{cases} 2^{2n-2} - (-1)^{n/2} 2^{n-1}, & n \text{ even} \\ 2^{2n-2} - 1, & n \text{ odd} \end{cases}$

Three features of this statement deserve emphasis.

The naive version is false, and the counterexample is instructive. "Bipartite, Hamiltonian path, at least one chord" does *not* imply the exponential branch, because it fails to exclude the cycle. The hexagon C_6 is the minimal witness: its single chord is the endpoint pair, of grade $N-2 = 10$, and two grade-10 monomials in 12 Majoranas must share at least eight indices, of which only the odd overlap 9 survives in a commutator, giving grade 2 only. The grade set is $\{2, 10\}$ and $\dim \mathfrak{g}_{C_6} = 132 = 2\binom{12}{2}$, not the 1056 that the naive statement would predict. C_8 gives 240. The 2×2 ladder is C_4 and therefore lies in the cycle branch, but at $N = 8$ the grades $\{2, 6\}$ happen to exhaust the whole class $k \equiv 2 \pmod 4$, so its dimension 56 coincides with the branching formula and hid the missing hypothesis.

In the branching branch the answer depends only on n. Neither the number of chords nor their positions enter. The open 2×5 ladder and the decagon with one non-endpoint chord both have $n = 10$ and both give exactly 262656; the 3×3 grid and the 3×3 grid minus an edge both give 65535. So the local-term algebra sees only the vertex count and the presence of branching. For odd n the volume element is the global X parity operator rather than a scalar; it is a nonidentity Pauli string that is not generated, which is the exact origin of the -1.

The trichotomy is the solvability dichotomy, made quantitative. The two low branches are precisely the free-fermion-solvable geometries: the open chain and the ring, i.e. the one-dimensional model and the two-dimensional Ising layer. Their algebras are *quadratic* in n and sit comfortably inside the Gaussian bound of Proposition 3.5, requiring only a linear number of modes. The branching branch is exponential and forces $\Theta(2^n)$ modes. So the boundary between solvable and unsolvable layer geometry appears as a dimension jump from $\Theta(n^2)$ to $\Theta(4^n)$, occurring exactly at $\Delta(\Gamma) \geq 3$ — the same threshold that appears in the Dolan–Grady defect (Theorem 3.1), the claw obstruction (Theorem 3.4), and the tridiagonal no-go (Theorem 3.3). Four independent mechanisms, one threshold.

Remark 4.10 (Necessity of bipartiteness; [COMPUTATION])**.** Dropping bipartiteness changes the answer rather than breaking the method: an odd chord distance contributes a grade $\equiv 0 \pmod 4$ and the closure grows to the full even-grade algebra. For C_5 with a chord, $n = 5$, the measured dimension is 510, against 255 for the $k \equiv 2 \pmod 4$ class and 510 for the full noncentral even algebra. Every simple-cubic layer graph is bipartite, so the physical case is the one covered by Theorem 4.9.

Scope, stated plainly: Theorem 4.9 is about the algebra generated by the local terms, over $\mathbb{Q}$, for graphs admitting a Hamiltonian path. It is not a statement about the spectrum of any single transfer matrix, and it does not exclude a Gaussian spectrum arising by some other route; the spectral obstructions of §5 are the complementary half of that question.

5 Basis-independent spectral obstructions

5.1 The Gaussian ceiling and the subset-product criterion

A Gaussian (free-fermion) operator with positive spectrum on m modes has spectrum

$$\Big\{\lambda_0 \prod_{k\in S} u_k \ :\ S \subseteq \{1,\dots,m\}\Big\}, \qquad u_k \geq 1, \tag{5}$$

a subset-product multiset. Two exact consequences are used.

Lemma 5.1 (Forced eigenvalue; [LEMMA])**.** *If the three lowest eigenvalues of* (5) *are distinct then* $\lambda_1\lambda_2/\lambda_0$ *is also an eigenvalue.*

Lemma 5.2 (Gaussian ceiling; [LEMMA])**.** *The number of distinct unordered pair products of* (5) *is at most* $3^n - 2^n$*. Within one parity half of a* $2m$*-Majorana Gaussian the corresponding maximum is* $A_{\mathrm{same}}(m) = \sum_{2\leq r\leq m,\ r\ \mathrm{even}} \binom{m}{r} 2^{m-r}$*, so* $A_{\mathrm{same}}(12) = 261625$*.*

Both are counting statements about (5) and involve no basis choice, which is what makes the resulting no-gos immune to "free fermions in disguise".

5.2 Inertia certificates

Theorem 5.3 (Theorem S; [THEOREM])**.** *At* $t = 1/3$*, equivalently* $q = e^{2K} = 5/3$*, the open* 2×3 *layer transfer operator is not a fermionic Gaussian operator on six modes in any Majorana basis. The three lowest eigenvalues are certified distinct with relative gaps* 53.0% *and* 40.9%*; the forced value* $\lambda_1\lambda_2/\lambda_0$ *lies in the exact rational window* $[0.0347241721739,\ 0.0347241728052]$*; and exact rational Sylvester inertia counts at the two outward-rounded endpoints are both* 8*, so the window contains no eigenvalue.*

The control chains of lengths four and six show the opposite inertia pattern, as expected for the two-dimensional free-fermion model. The asymmetry is essential and is stated in the artifact: equal inertia counts prove *absence*; unequal counts show only that some eigenvalue lies in the window, never that the forced value is attained. Theorem S extends to $t = 1/5, 2/5, 1/2$ on 2×3 and to the 2×4 layer at $t = 1/3$ through certified symmetry sectors. A companion theorem for the two physical $P = \prod_i X_i$ sectors shows they are not the even/odd halves of any single six-mode Gaussian, in either assignment.

5.3 The pair-product invariant

The inertia route needs an ordering and a positivity hypothesis. The pair-product route needs neither: count distinct unordered products $\lambda_i\lambda_j$ and compare with Lemma 5.2. Concretely, if $C(z) = \prod_{i<j}(z - \lambda_i\lambda_j)$ then the number of distinct products is $\binom{N}{2} - \deg\gcd(C, C')$, and after monic integral normalisation every modular gcd degree is an upper bound for the characteristic-zero one, which is the direction that yields a lower bound on the count.

Theorem 5.4 (Ordering-free no-gos; [THEOREM])**.** *With* N *the block dimension, at the stated couplings and at both primes* 1000003 *and* 2000003*:*

layer	N	deg gcd	*distinct products* $\geq$	*Gaussian ceiling*
2×3 $(t = 1/3)$	64	385	1631	665
2×4 $(t = 1/3)$	256	9329	23311	6305
3×3 $(t = 1/3)$	512	59641	71175	19171
2×5 $(t = 1/3)$	1024	124921	398855	58025

*In each case the spectrum is not a full n-mode Gaussian subset-product multiset. Open-chain controls of the same size hit their ceilings exactly (*1351, 465751, 111645 *respectively), and characteristic-zero equality with the measured* gcd *degrees is* not *claimed.*

By Lemma 2.2 these point certificates upgrade: the pair polynomial is monic in z over $\mathbb{Z}[t]$ after the scalar $(2t)^3(1+t^2)^4$, so for every positive real t outside an explicit finite exceptional set the conclusion persists. The exceptional-set bounds are 422472960 for 2×3 and 51576065587200 for 2×5; the parity-half version has bound 25534083 including the scalar zeros $0, \pm i$.

5.4 The 2×6 certificate

The largest layer certified so far is the open 2×6 ladder, on twelve sites.

Theorem 5.5 (Theorem 2×6-NG; [THEOREM])**.** *At $t = 1/3$ the physical $P = +1$ sector of the open 2×6 layer is not either parity half of any twelve-mode Gaussian subset-product multiset. The physical $(P = +1, \rho = +1)$ block has dimension* 1056*, hence $\binom{1056}{2} = 557040$ pair slots, and $557040 - A_{\mathrm{same}}(12) = 295415$. On both primes* 1000003 *and* 2000003 *the exact pair-polynomial computation gives*

$$\deg \gcd(C, C') \;\leq\; 295414 \;<\; 295415,$$

so the block has at least 261626 *distinct within-half pair products, at least one above the Gaussian ceiling $A_{\mathrm{same}}(12) = 261625$.*

The margin is one. That is a razor, so the certificate is worded strictly one-sidedly: the computation proves "at least one above", never "exactly one". Two independent safeguards were required before the statement was accepted. First, the Euclidean remainder sequence was checkpointed and the final certified tail, 25 steps, was replayed by a separate list-schoolbook implementation with bit-identical remainders. Second, a standalone verifier performs 38 checks that re-derive the result from the stored state without trusting any conclusion field: content-key and digest chains, step accounting $261600 + 25 = 261625$, the exit remainder against the artifact digest, and the two-implementation remainder equality at both primes.

5.5 All-size theorems and the isotropic gap

Theorem 5.6 (Theorem F, all-size anisotropic; [THEOREM])**.** *Let G be a finite simple graph with a vertex of degree at least three and $n = |V| \geq 4$. For the inhomogeneous rational layer family with independent site parameters t_v and bond parameters w_e, there is a nonzero polynomial ρ_G with*

$$\deg \rho_G \;\leq\; 4(2n + |E|)S(S-1), \qquad S = \binom{2^n}{2},$$

such that whenever $\rho_G \neq 0$ the spectrum has at least $3^n - 2^n + 48 \cdot 3^{n-4}$ distinct unordered slot-pair products. Since a full n-mode Gaussian multiset has at most $3^n - 2^n$, the generic inhomogeneous layer is not Gaussian, with an excess that is the constant fraction $16/27$ of 3^n.

The mechanism is a ceiling lemma, an exact tensor identity $r = (3^m - 2^m)\, r_{\mathrm{all}}(\mathrm{core}) + 2^m\, r(\mathrm{core})$, a claw seed with $r_{\mathrm{all}} \geq 129 > 81$, and monic specialization. The exceptional set is a proper algebraic hypersurface, hence Lebesgue-null.

Remark 5.7 (The isotropic gap: a split verdict; [THEOREM]/[UNRESOLVED])**.** Theorem 5.6 is genuinely anisotropic, and the two halves of the isotropic repair now have opposite verdicts. The *core* half succeeds: at $t = 1/3$, at both primes, three isotropic cores are certified strictly above their own Gaussian ceiling — the paw ($r_{\mathbb{Q}} \geq 119 > 65$), the five-site branch-length-$(2,1,1)$ tree ($417 > 211$), and the open 2×2 grid with a pendant ($475 > 211$) — while the bare claw only reaches its ceiling ($65 = 65$) and the star $K_{1,4}$ falls below it ($147 < 211$), which is exactly why the bare claw seed never sufficed. By the monic upgrade each of the three is non-Gaussian at all but finitely many isotropic couplings, with exceptional-degree bounds 913920, 17677440, 19641600. The *decoupling* half fails, and provably so: the graph-piece tensor factorization is not an identity on the isotropic curve, since the crossing-factor minor is $q(t)^{2c} - 1$ with cleared numerator $(1+t^2)^{2c} - (2t)^{2c} > 0$ for $0 < t < 1$; and the natural localizing family is too small, since $r(K_{1,3} \sqcup mK_1) \leq 136(2m+1) < 3^{m+4} - 2^{m+4}$ for every $m \geq 3$, i.e. linear growth against an exponential ceiling. So no isotropic all-size *spectral* theorem is claimed; what is now known is that any proof must be non-localizing. Note that the isotropic all-size *algebraic* statement is obtained independently in §4.3.

A separate exact analysis settles what happens at the boundary of an absence interval: the forced value crosses the spectrum at real algebraic couplings, four exact crossing brackets are isolated, each containing exactly one crossing by five-variable Krawczyk contraction, and absence is proved on three connected crossing-free rational intervals, the widest of width 1/1048576. Global crossing exhaustiveness remains open.

6 Tensor-network and planarity ansatzes

6.1 Tetrahedron and RLLL equations

Three-dimensional integrability in the Zamolodchikov sense requires a tetrahedron equation [33, 34, 3]. With $L(q)$ the verified six-leg Ising tensor,

$$R_{123}L_{145}(q)L_{246}(q)L_{356}(q) = L_{356}(q)L_{246}(q)L_{145}(q)R_{123}.$$

Theorem 6.1 (Ungraded tetrahedron no-go; [THEOREM])**.** *Over every field of characteristic zero, if $q \notin \{0, 1, -1\}$ then the only 8×8 solution R of the identical-L equation is $R = 0$. In particular no invertible auxiliary R exists for finite ferromagnetic weights $0 < q < 1$.*

The proof concatenates two complementary 64×64 minors, $2q^{109}(q-1)^{92}(q+1)^{77}(q^3+2q-1)$ and $2q^{110}(q-1)^{92}(q+1)^{77}(q^2+1)$, whose remaining factors are coprime by an explicit Bézout identity. The graded case is even sharper: every $\mathbb{Z}_2$-grade-preserving auxiliary vanishes for all q outside $\{0, \pm 1\}$ because $\det B(q) = -2q^{56}(q-1)^{45}(q+1)^{39}$. For the canonical one-line-cabled 16384×256 system the characteristic-zero rank-drop locus is exactly $\{-1, 0, 1\}$, with complementary-minor gcd $q^{584}(q-1)^{580}(q+1)^{546}(q^2+1)^4$ and $q = \pm i$ cleared as spurious. Scope: identical local factors, this vertex-type equation, auxiliary dimension eight (sixteen when cabled). Higher auxiliary dimensions, nonidentical factors, spectral-parameter families, and dynamical relations are not excluded.

6.2 Kac–Ward

The two-dimensional solution has a combinatorial form: a determinant over directed edges [19]. A full translation-invariant scalar three-dimensional analogue has 30 directed-direction weights, reduced to 25 variables on a generic five-edge gauge chart. The exact two-dimensional identity is reproduced as a control.

Theorem 6.2 (Positive-real scalar no-go; [THEOREM]). *No assignment of nonnegative real values to the* 30 *allowed direction transitions satisfies the coefficientwise scalar Kac–Ward identity on every free cubic box. The order-four trace equation on the free* $2 \times 2 \times 2$ *box is*

$$F_4 = 48 + 8\big(M_{xy,+} + M_{xy,-} + M_{xz,+} + M_{xz,-} + M_{yz,+} + M_{yz,-}\big),$$

strictly positive on the nonnegative orthant, and a stored rational Positivstellensatz identity certifies infeasibility on all 32 *selected gauge strata.*

Theorem 6.3 (Rational branch map; [THEOREM]/[UNRESOLVED]). *Of the* 32 *sign branches of the rational family,* 31 *are empty over* $\mathbb{Q}$*, certified by thin-box integer constants; branch* 00000 *is empty by the exact constant* $2F_{(3\times3\times2,4)} - 3F_{(2\times3\times4,4)} = 56$*. Branch* 11111 *is open. It has a nonsingular* $\mathbb{F}_5$ *point with* $\det J = 2$*, hence a genuine* $\mathbb{Q}_5$ *component and a proper rational ideal, so* no *Nullstellensatz emptiness certificate exists at any multiplier degree. Its local Hensel lift fails an independent* $3 \times 3 \times 3$ *holdout at orders* $8, 10, 12$ *modulo* 625*, with residues* $[250, 55, 266]$*,* $[456, 350, 132]$*,* $[430, 480, 286]$*.*

Deciding branch 11111 was first scoped honestly rather than claimed: an initial thin-torus census fitted only 1 of 16384 sections into a 20-second process-time cap, projecting about 100002.627584 process seconds, and its artifact was labelled `PREFLIGHT`/[UNRESOLVED], because a measured timeout is not a theorem. Two later results replaced that projection.

Theorem 6.4 (No orbit reduction exists; [THEOREM]). *On the diagonal-normalised* 13*-coordinate branch-*11111 *thin chart, the rank-three torus stabiliser and the gauge-tree residual are trivial, and although the* 14*-shape construction catalogue has a point group of order* 16*, every nonidentity element fails to induce a map on the six-solve seven-held fibration, with paired zero witnesses on the construction variety itself. Hence the sound section group is trivial: there are exactly* 16384 *singleton orbits and no quotient reduction is available.*

That closes the orbit-quotient line of attack negatively, which is useful precisely because it stops future work from re-attempting it. The census was then completed outright by exact character evaluation over all $4^{13} = 67108864$ unit points: there are exactly 2960 construction zeros lying in 2437 sections, and every one has an exact disposition — 2232 killed by a nonzero order-8 residue mod 5, 664 by a nonzero order-12 residue mod 5, 45 unique nonsingular lifts by a nonzero order-8 residue mod 625, and 19 singular points by exhaustive affine lift obstruction through mod 25, 125 or 625. The resulting classification is 0 sections `EMPTY_OVER_Q`, 2437 `HAS_LOCAL_POINT_BUT_FAILS_HOLDOUT`, and 13947 `UNDECIDED`. The scope sentence matters more than the counts: for each of the 16384 slices with held coordinates fixed in $\{1, 2, 3, 4\}^7$, no $\mathbb{Q}_5$ point with all six solve coordinates 5-adic units satisfies both the 42 construction equations and the recorded holdouts. Higher 5-adic digits, non-unit valuations, zero reductions, chart-degenerate loci, other primes, and arbitrary characteristic-zero points are all untouched. Branch 11111 remains open over $\mathbb{Q}$.

7 The exact computational base

7.1 Series

The reduced free energy is computed by exact finite-lattice methods with integer states, profile injections, orbit normalisation, and per-box Chinese-remainder uniqueness certificates.

Theorem 7.1 (High-temperature coefficient; [THEOREM]/[COMPUTATION])**.** *Through order* 28 *only bounding boxes with* $a+b+c \le 17$ *contribute, an inventory of* 147 *canonical permutation classes, and*

$$[v^{28}]\phi(v) = \frac{2102198327465307}{28}, \qquad a_{28} = \frac{525549581866326}{7},$$

with $[v^{27}]\phi = [v^{25}]\phi = 0$ *and every coefficient through* v^{26} *reproducing the independent predecessor artifact, including* $[v^{26}]\phi = 115377512914251/26$ *and* $[v^{24}]\phi = 2135670379057/8$.

Theorem 7.2 (Low-temperature coefficients; [THEOREM]/[COMPUTATION])**.** *With* $x = e^{-2K}$*, the order-*56 *inventory has* $a+b+c \le 15$*, i.e.* 455 *ordered or* 102 *permutation classes, and is complete through* x^{56}*; by the same surface inequality the first incomplete inventory is at* x^{58}*. The exact endpoint values are*

$$[x^{56}]\phi = -\frac{979227570369}{4}, \qquad [x^{54}]\phi = \frac{675410878105}{9}, \qquad [x^{52}]\phi = -\frac{46131904167}{2},$$

all odd coefficients vanish, and an independent fresh-prime audit reproduces the result.

Both sides are witnessed after the fact by external tables [2, 1, 17, 4]; those tables are validation witnesses and producer gates, never computational inputs.

7.2 Structure frontiers

Exact coefficients permit exact refutation of closed-form ansatzes. A grid of 633 admissible cells over algebraic, first-order differential-algebraic, Euler-ODE and Mahler spaces gives 322 rows with no relation at budget, each with a stored exact nonzero minor, 159 rows refuted by an exact holdout residual, and 152 surviving rows, every one certified truncation-unobservable, so there are zero unexplained survivors. The one high-temperature Euler-ODE survivor was refuted at its first unobservable order with exact residual

$$\frac{4997524512213726102393494613686474327029458488 08}{11} \neq 0 \qquad \text{at } v^{24},$$

with v^{26} and v^{28} recorded as confirmation only. This is a finite-prefix, finite-ansatz negative result: it is not a non-D-finiteness theorem, and it is stated as such.

7.3 Certified critical interval

Theorem 7.3 (Rigorous interval; [THEOREM])**.**

$$0.21221597532702316272675171742785778060 20 \le K_c \le 0.2527310098586630030260020266135701299926.$$

The lower endpoint comes from a self-avoiding-walk route: the all-n first-backtrack inequality $\mu^n \le c_n - a_{n-1}$, combined with an exact union certificate $a_{35} \ge 197246546107018625783 5$ obtained by inclusion–exclusion over a 404-member two/three-block height-schedule family with exact overlap correction 503020433441526165912 0 against the naive disjoint volume 700266979548544791695 5. Exact enumeration data are external [27, 23], and the chaining is all-n [21] while the union is a finite a_{35} certificate. The upper endpoint is an audited incumbent from a separate rigorous route; two attempts to improve it were closed exactly:

Theorem 7.4 (Peierls limitation; [THEOREM]). *The plain Peierls head-plus-tail contour certificate cannot beat the incumbent. With the exact cell-class census through $n \le 10$ (*9,638,143 *classes flood-filled, none cavity-bearing, the smallest cavity needing exactly* 11 *cells) the rooted contour counts extend to $N(26) = 20448$ and $N(28) = 33240$, the certified partial sum already exceeds $1/2$, the crossing $K^* = 0.2558326$ sits 0.003102 above the incumbent by three-way bisection, and the proved tail mass $1.65288 > 1/2$ annihilates the head-only apparent gain.*

Separately, the audited two-point infrared/GKS class is exactly optimal at the incumbent, with an $O(1/L)$ torus linear-programming floor, so any improvement must use genuinely four-point or paired-momentum information. The classical Peierls argument [25, 16, 15] is thus method-limited here, not merely unoptimised.

Two further certificate classes were named precisely and then exactly refuted, at the common rational challenge $v = 6/25$, i.e. $K^\dagger = \operatorname{atanh}(6/25)$, certified more than 0.00795689719931011 below the incumbent. In the aggregate one-moment paired-momentum class the rational two-variable 2×2-moment semidefinite program has exact optimum $\sup s = 1$ and uniform order-floor optimum $\eta^* = 0$ for every $K \le I_3/2$, so the fixed challenge $\eta = 1/1024$ is exactly infeasible with separating gap $1/1024$. In the single-edge conditional-domination random-current class the exact linear-programming optimum is $p^* = v^2$, which at $v = 6/25$ is $36/625$ against a self-proved planar threshold $5/6$, short by exactly $2909/3750$, and uniformly $p^* < 1/16$ for every $K \le I_3/2$. As a by-product the C_4 plaquette double-current enumeration gives $P(0 \leftrightarrow 2) = \langle \sigma_0 \sigma_2 \rangle^2$ exactly. Neither class excludes mode-resolved four-point or DLR constraints, multi-edge or block current certificates, or sourced-current inequalities.

7.4 Controls, and falsification of published claims

Three controls run continuously. Independent exact enumeration and transfer propagation agree on 17 lattice and boundary cases, and the layer transfer operator reproduces exact three-dimensional partition functions on seven lattices. In two dimensions, Kaufman's finite-torus formula [20] agrees with exact integer transfer matrices in eight cases to maximum relative error $3.63867276317 \cdot 10^{-60}$, and the parity-resolved logarithm has maximum off-Majorana-bilinear coefficient $1.246 \cdot 10^{-15}$. Exact finite-volume Ising–$\mathbb{Z}_2$ gauge duality [30] passes 27 integer checks, and the simple-cubic model is not self-dual. Lee–Yang zeros [22, 32] pass unit-circle checks on 13 lattices through $4 \times 4 \times 4$, with largest stored 64-site radial residual $9.1044993726259 \cdot 10^{-92}$, and exact $5 \times 4 \times 4$ and $5 \times 5 \times 4$ zeros are isolated to width 2^{-260}; no thermodynamic edge exponent is reported, because at accessible sizes it is provably non-identifiable.

Claimed exact solutions were tested rather than dismissed. The claims of [36, 35] fail independent critical, series and symmetry tests; both evaluable claimed free energies have logarithmic rather than the externally established critical behaviour, consistent with the published objections [31, 26].

8 Verification methodology and corrections

8.1 Why the numbers should be believed

Every claim above is backed by an artifact and a verifier that re-derives the decisive quantity independently. The current acceptance gate is 123 standalone verifiers with `FINAL: 123 total, 123 passed, 0 failed`. Verifiers must fail, not warn, when a row of a universal claim is uncovered. An adversarial replication script deliberately prints `FAIL` for two historical over-general claims while exiting zero, so that the corrected scopes stay visible.

8.2 Withdrawn and corrected claims

Listing these is part of the result.

- **Theorem U2: withdrawn, then restored.** A modular lower bound was once substituted into an upper bound, so the counts 3893 and 35597 were withdrawn. They are now *proved*. Exact characteristic-zero computation on the five-site tree T with edges $\{01, 02, 03, 34\}$ gives $r(T) = 417$ and $r_{\rm all}(T) = 445$ at all four named physical points $(t, w) = (1/3, 5/3), (1/4, 17/8), (2/5, 29/20), (1/5, 13/5)$ — equivalently, the degree-496 and degree-528 pair polynomials have exact $\mathbb{Q}$ derivative-gcd degrees 79 and 83. The exact core upper bounds then squeeze the modular lower bounds, certifying $r(T \sqcup P_2) = 3893$, $r_{\rm all}(T \sqcup P_2) = 4005$ and $r(T \sqcup P_4) = 35597$, with decoupled excess $202 \cdot 3^m + 4 \cdot 2^m$. Consequently the hypothesis CZ of Theorem F2 is discharged and F2 is unconditional for $m = 2, 4$. The hypothesis INJ$(n - 5)$ for other chain lengths remains open, as does whether the fully isotropic point lies on the exceptional hypersurface. This entry is kept in the corrections list deliberately: the original error and its repair are both part of the record.
- **Dolan–Grady zero set corrected.** "Dolan–Grady iff 2-regular" is false; the exact condition is $d(v) \in \{0, 2\}$ for every vertex.
- **Tridiagonal witness corrected.** A universal one-row witness does not exist; some geometries need the stored four-row certificate.
- **Flux theorem scope narrowed.** The proved centralizer is within the Pauli group for the individual generators, not the full commutant of the sums; translation on a triangle is a counterexample to the stronger reading.
- **Kac–Ward scope narrowed.** Ruling out the strict two-weight slice never decided the 30-weight family; the current status is Theorems 6.2 and 6.3.
- **$\mathfrak{sp}_{14}$-type labels withdrawn** where saturation failed, and the earlier Davies-quotient argument was retracted as self-contradictory before being replaced by Corollary 4.2.
- **SectorSaturationCyclicity refuted**, which removed the single missing step of an earlier conditional ladder theorem: $W_L = 0, 2, 10, 66, 364, 1822$ at $L = 3, \ldots, 8$ with cyclic dimensions 14/14, 42/44, 142/152, 494/560, 1780/2144.
- **Watson normalisation corrected**: the gamma denominator is $32\pi^3$, not $4\pi^3$.
- **Alternation-law quantifiers downgraded on review**: containment is all-L, equality is $L \leq 5$; the intermediate stronger claims were withdrawn in the same session they were made.

9 Open problems

In descending order of value-times-tractability as judged from the current evidence.

1. **The general-graph grade classification.** The $2 \times L$ theorem has a Clifford-grade explanation that appears to extend to every bipartite layer graph with a Hamiltonian path, with a trichotomy between paths, even cycles, and branching graphs. Settle the general statement and its exact hypotheses; the 3×3 closure $65535 = 2^{16} - 1$ is already an instance.
2. **A proved law for W_L.** A closed form, recurrence, or one-sided bound $W_L \leq (1 - c)K_L$ would convert the certified finite bounds $\dim\langle A_L, B_L\rangle \geq K_L - W_L$ into an all-L theorem and hence, with Proposition 3.5, into an unconditional isotropic all-L free-fermion no-go for ladder transfer generators.

3. **The isotropic gap in Theorem 5.6.** Find an isotropic core certified above the ceiling at uniform fields together with an isotropic decoupling whose remainder keeps modes distinct.
4. **Exact characteristic-zero $r(T)$ and $r_{\text{all}}(T)$**, which would discharge one hypothesis of Theorem F2 and reinstate a corrected form of the withdrawn Theorem U2.
5. **Kac–Ward branch** 11111 over $\mathbb{Q}$ and $\mathbb{C}$: prove a sound orbit quotient, then run the reduced census. Emptiness cannot be certified by Nullstellensatz at any degree, so the decision must be structural or by exhaustive holdout.
6. **Upper endpoint below the incumbent**, necessarily using four-point or paired-momentum constraints, or random-current inequalities evaluated exactly on finite boxes.
7. **Longer series**: high-temperature v^{30}, low-temperature x^{60} and x^{62}, the last needing a 30-spin cross-section. All are resource-limited, not mathematically blocked.
8. **Larger layers**: the 3×4 pair-product certificate, the open 4×4 algebra beyond dim ≥ 1794, and the 3×3 local-term algebra, which sits outside the $2 \times L$ family.

Artifact map and reproduction

Every statement above corresponds to a producer script, an artifact, a standalone verifier, and a tagged note in the accompanying bundle. Principal entry points:

- Local algebra: `proofs/dolan_grady_defect.md`, `proofs/tridiagonal_nogo.md`, `proofs/algebraic_obstruction.md`, `proofs/dg_simultaneous.md`, `proofs/local_commutant.md`, `proofs/extensive_3d_s5.md`.
- Characteristic-zero structure: `proofs/char0_structure.md`, `proofs/char0_complete_2x4.md`, `proofs/char0_3x3.md`, `proofs/onsager_quotient_nogo.md`, `proofs/alternation_law.md`, `proofs/alternation_l5_char0.md`, `proofs/char0_4x4.md`.
- Spectral: `proofs/spectral_gaussianity.md`, `proofs/pair_product_obstruction.md`, `proofs/pair_product_scale.md`, `proofs/pair_product_3x3.md`, `proofs/pair_product_2x5.md`, `proofs/gaussian_2x6.md`, `proofs/parity_pair_product.md`, `proofs/allsize_gaussian.md`, `proofs/spectral_interval2.md`.
- Tensor and planarity: `proofs/tetra_ungraded.md`, `proofs/tetra16_locus.md`, `proofs/kac_ward_positive_real.md`, `proofs/kac_ward_full_family.md`, `proofs/kw_cubic.md`, `results/kac_ward/full_thin_census.json`.
- Series and bounds: `proofs/ht_v28.md`, `proofs/lt_x56.md`, `proofs/series_frontier3.md`, `proofs/saw_union4.md`, `proofs/kc_bounds_improved.md`, `proofs/upper_infrared.md`, `proofs/upper_beyond.md`, `proofs/mag_floor.md`.
- Orientation: `problem_specification.md`, `notes/CODE_API.md`, `README.md`, `checkpoints/verified_results.json`, `checkpoints/current_state.md`, `checkpoints/next_actions.md`, `notes/hypotheses.csv`.

Run every verifier from the repository root with `.venv/bin/python`. Producers overwrite their own JSON artifacts and refresh timestamps; the acceptance suite is an integration gate and does not replace the targeted standalone checks. Do not substitute the benchmark critical coupling for any exact input.

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